Course Reminders

Due Sunday (11:59 PM)

- A3
- Q4
- Project Proposal <- accept repo invite (resent; expires in 7d); talk with your group!

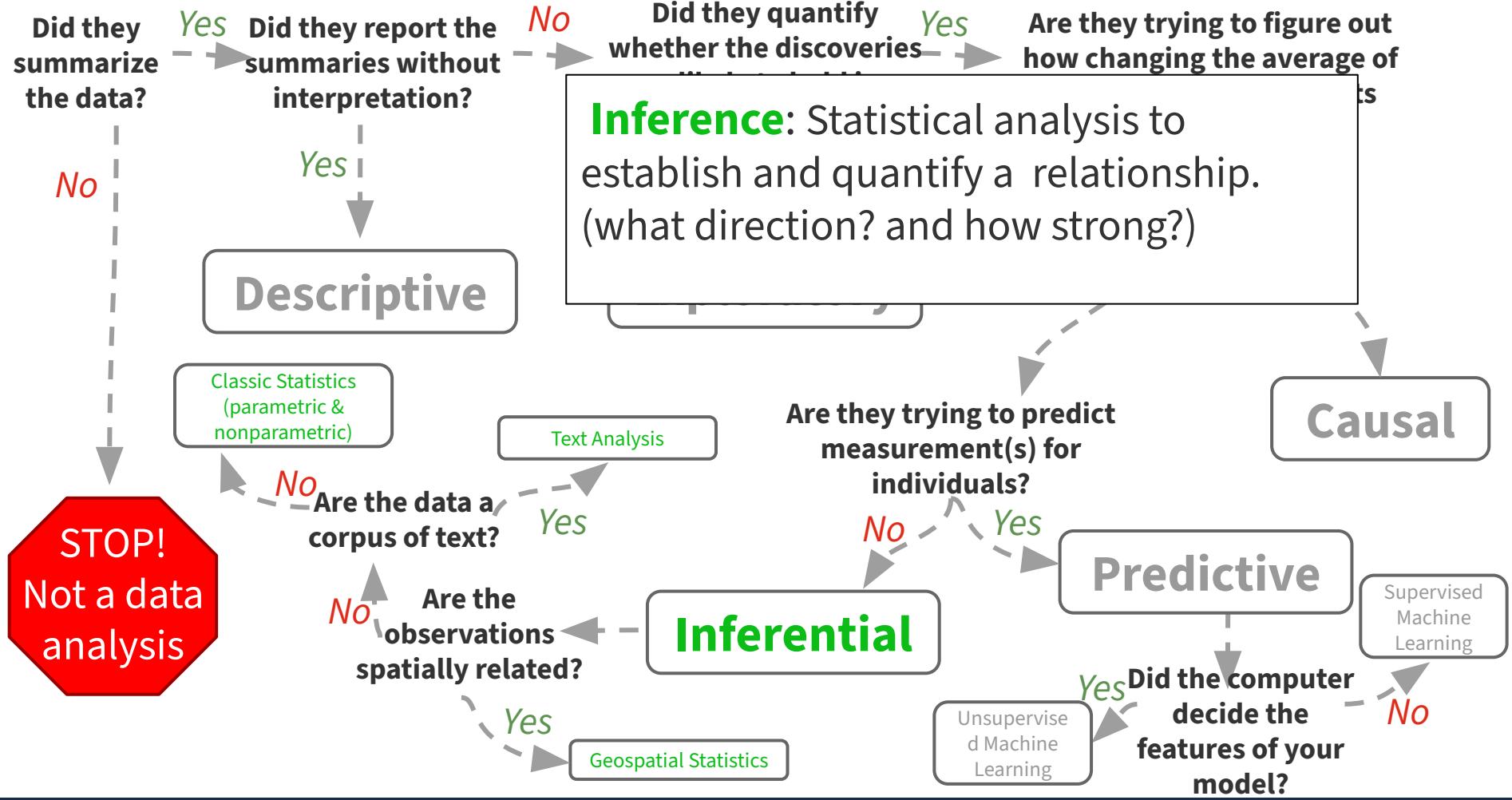
Notes:

- Inference I (slides + notebook) material will be on *A4* (released tonight; due *next* Sunday)
- Discussion attendance: it's there for people who *attend/participate*….not just show up to complete the form

Inference I:
Comparing Groups

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UC San Diego

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Did they summarize the data?
Yes
Did they report the summaries without interpretation?
No
Did they quantify whether the discoveries
Yes
Are they trying to figure out how changing the average of
Inference: Statistical analysis to establish and quantify a relationship. (what direction? and how strong?)
No
Yes
Descriptive
Classic Statistics (parametric & nonparametric)
Text Analysis
Are they trying to predict measurement(s) for individuals?
Causal
No
Are the data a corpus of text?
Yes
STOP! Not a data analysis
No
Yes
Predictive
No
Are the observations spatially related?
Inferential
Supervised Machine Learning
Yes
Did the computer decide the features of your model?
No
Yes
Unsupervised Machine Learning
Geospatial Statistics



We want to learn something about this...

Sampling

Inference

....but we can only *actually* collect data from this



Sample

1million comments from 2020

CC licensed clip art from pixabay.com

Air pollution control

??

Lifespan

What is the relationship between air pollution control and lifespan?

NIH Public Access
Author Manuscript
Epidemiology. Author manuscript; available in PMC 2014 January 01.

Published in final edited form as:
Epidemiology. 2013 January ; 24(1): 23–31. doi:10.1097/EDE.0b013e3182770237.

The Effect of Air Pollution Control on Life Expectancy in the United States: An Analysis of 545 US counties for the period 2000 to 2007

Andrew W. Correia,
Department of Biostatistics, Harvard School of Public Health, 655 Huntington Avenue, HSPH Building 2, 4th Floor, Boston, MA 02115

C. Arden Pope III,
Department of Economics, Brigham Young University, 142 Faculty Office Building, Provo, UT 84602

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Yun Wang,
Department of Biostatistics, Harvard School of Public Health, 655 Huntington Avenue, HSPH Building 2, 4th Floor, Boston, MA 02115

Majid Ezzati, and
MRC-HPA Centre for Environment and Health and Department of Epidemiology and Biostatistics, Imperial College London, Norfolk Place, St Mary's Campus, London W2 1PG

Francesca Dominici
Department of Biostatistics, Harvard School of Public Health, 655 Huntington Avenue, HSPH Building 2, 4th Floor, Boston, MA 02115, fdominic@hsph.harvard.edu, P: (617) 432-1056; F: (617)-739-1781

A decrease of 10 µg/m3 in the concentration of $PM_{2.5}$ was associated with an increase in mean life expectancy of 0.35 years SD= 0.16 years, p = 0.033). This association was stronger in more urban and densely populated counties.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3521092/

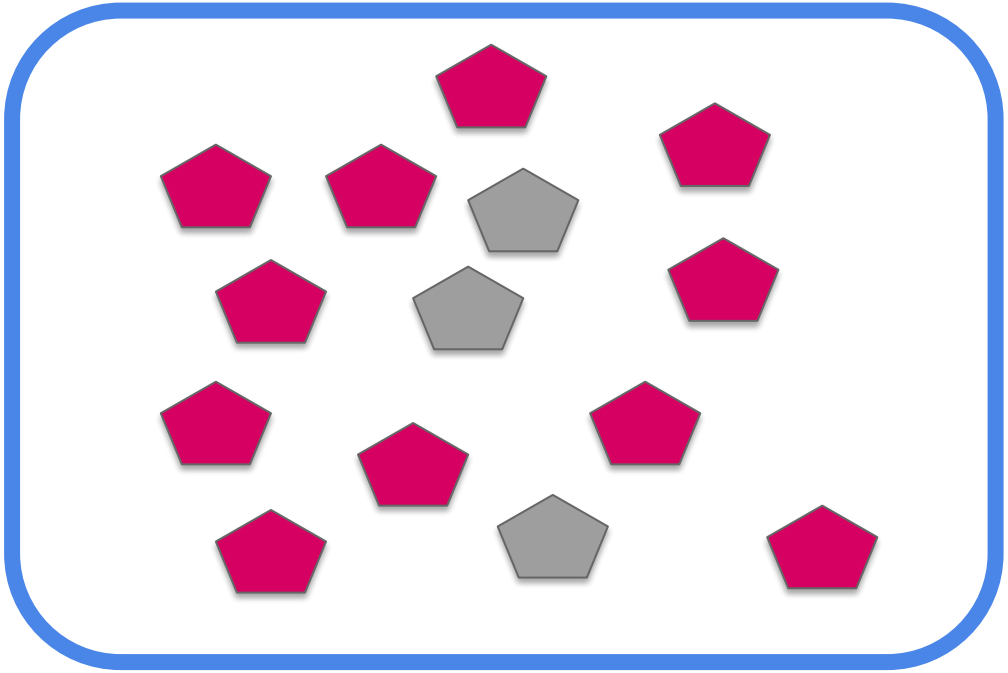
Establishing & Stating Your Null and Alternative Hypotheses Helps Guide Your Analysis

Null Hypothesis:

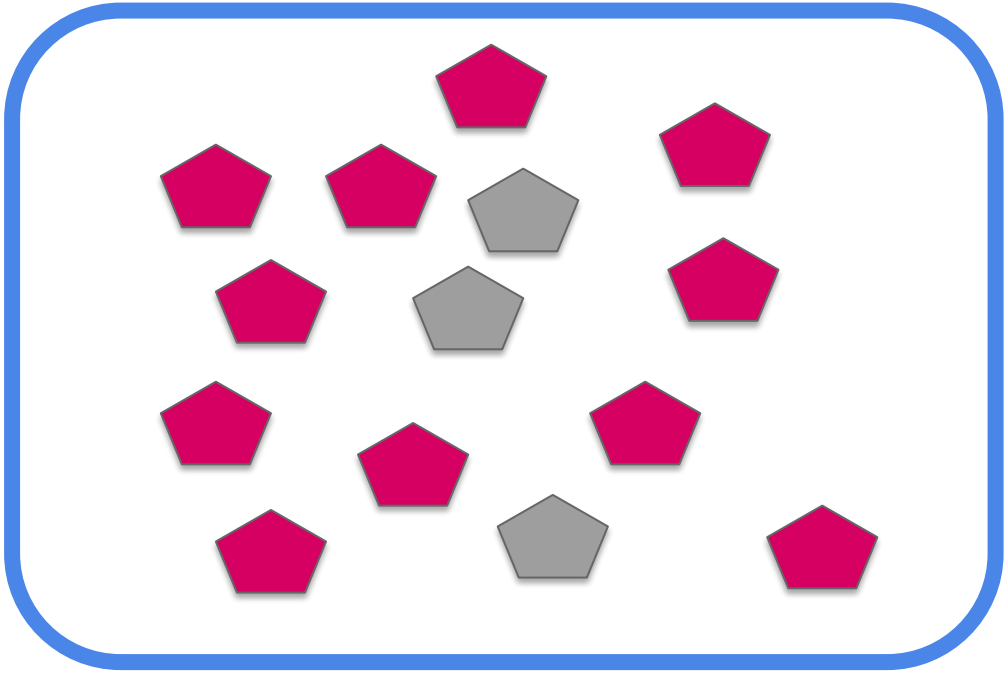
H_0: Air pollution has no effect on lifespan

Alternative Hypothesis:

H_a: Air pollution has an effect on lifespan

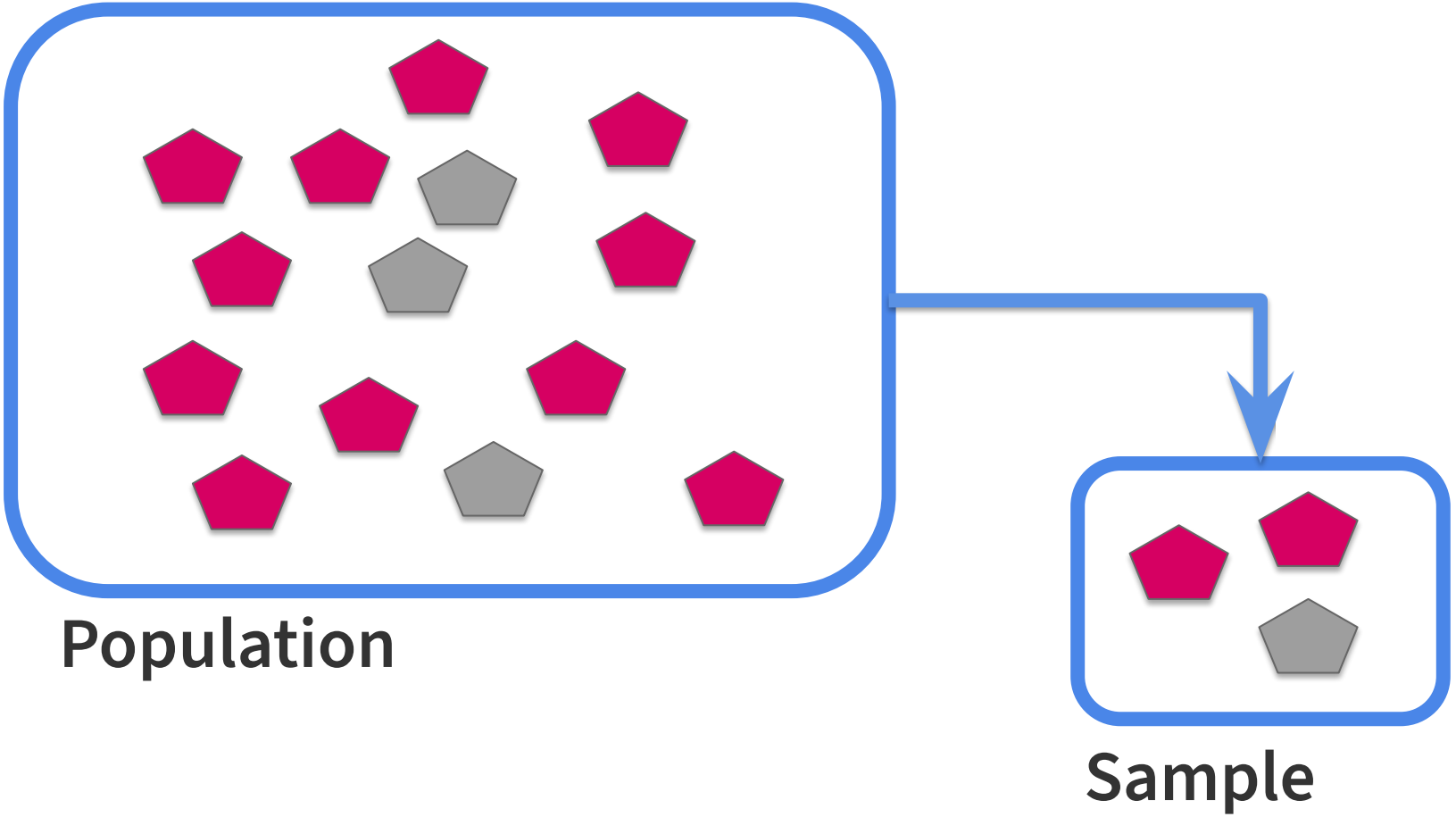


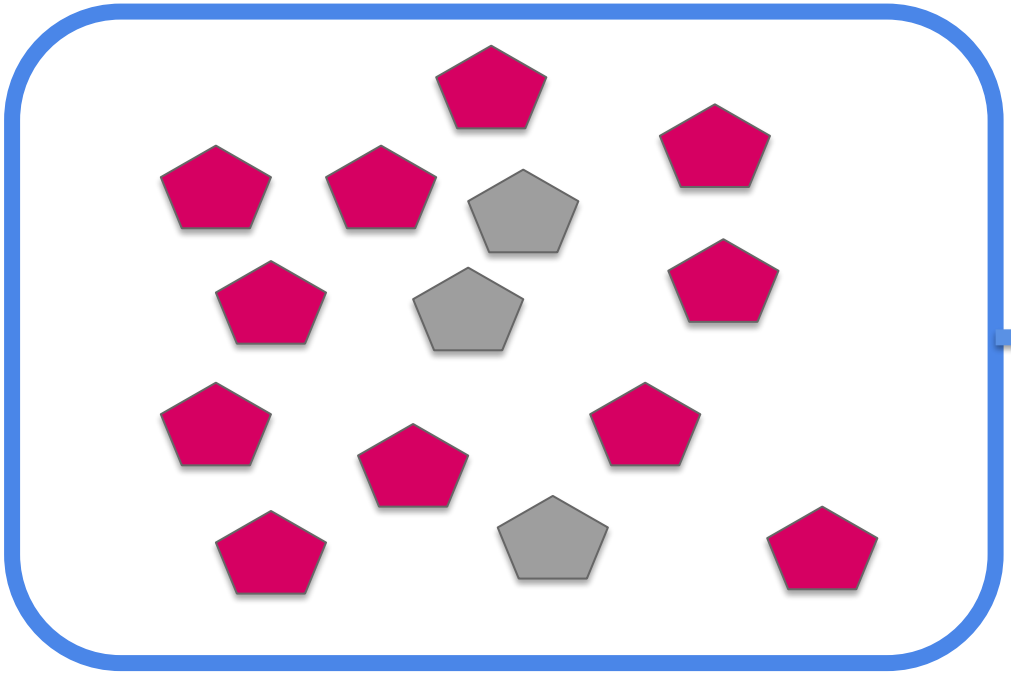
Population



Population

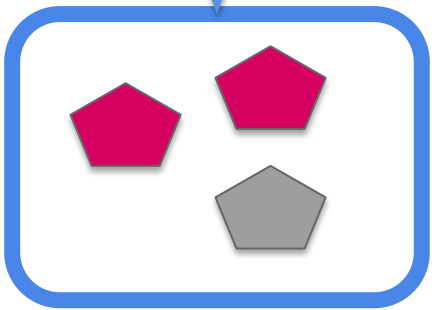
In our air pollution question, the population would be every individual in the US


Population
Sample

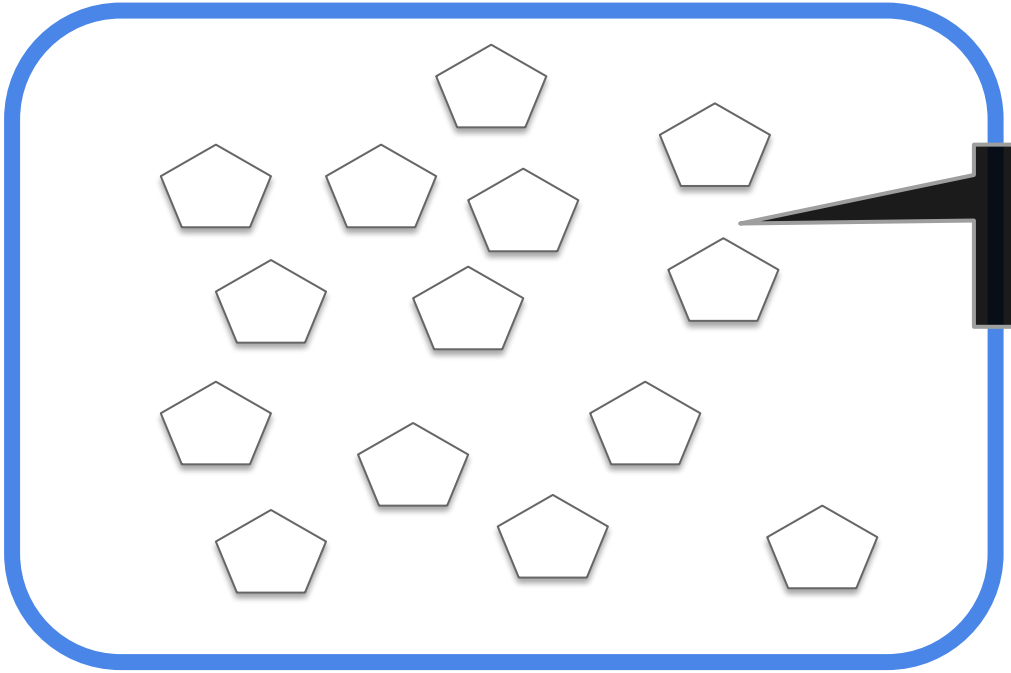


Population

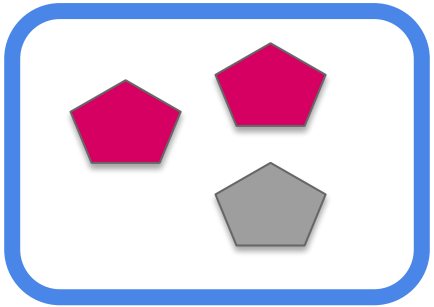
In our air pollution example, the sample would be measurements for some of the US



Sample

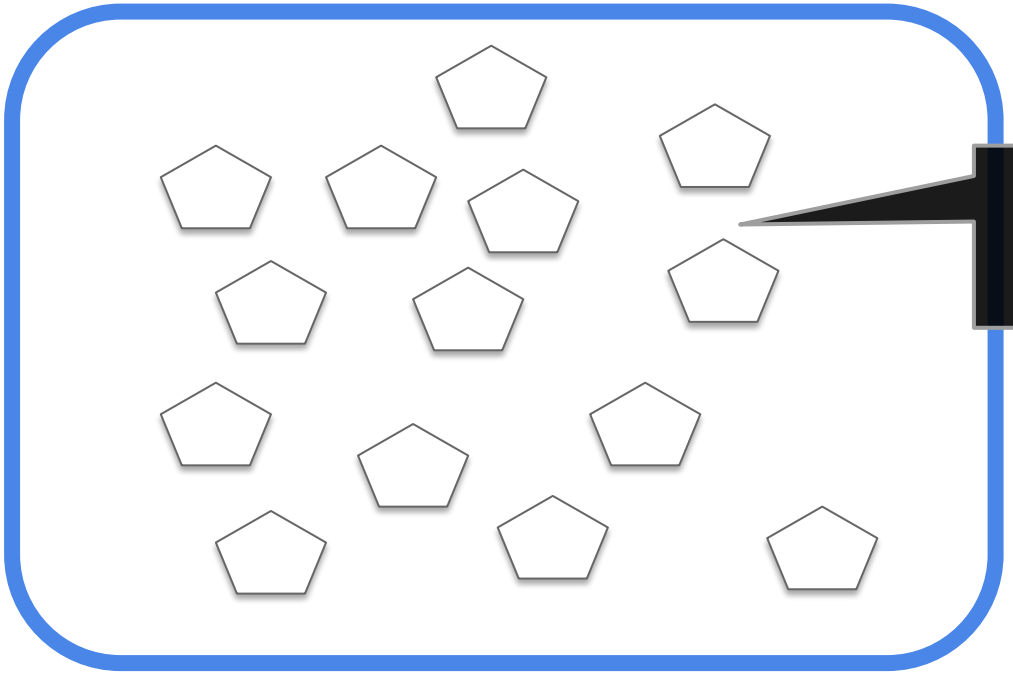


Population

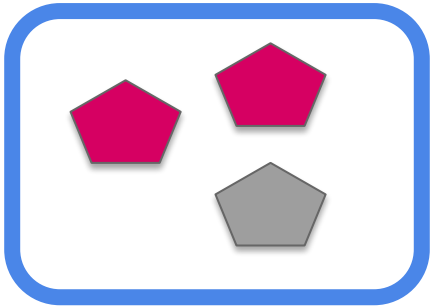


Sample

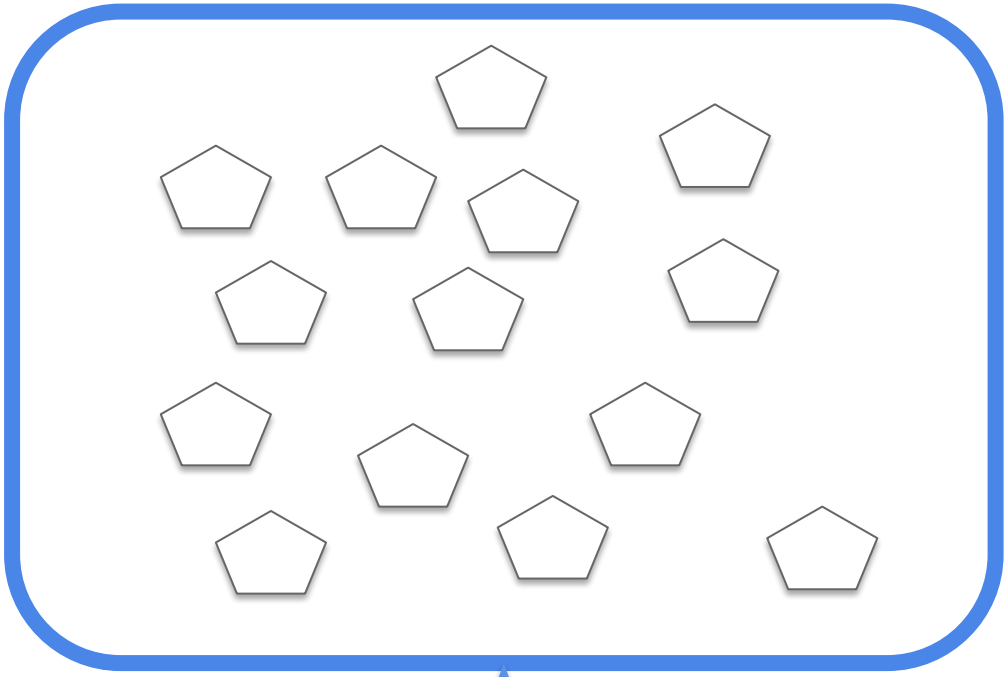
We don’t know how much air pollution each individual is exposed to.



Population

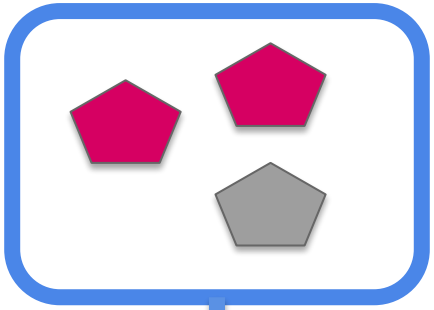


Sample



Population

Based on the relationship we see in our sample, we can <u>infer</u> the answer to our question in our population



Sample

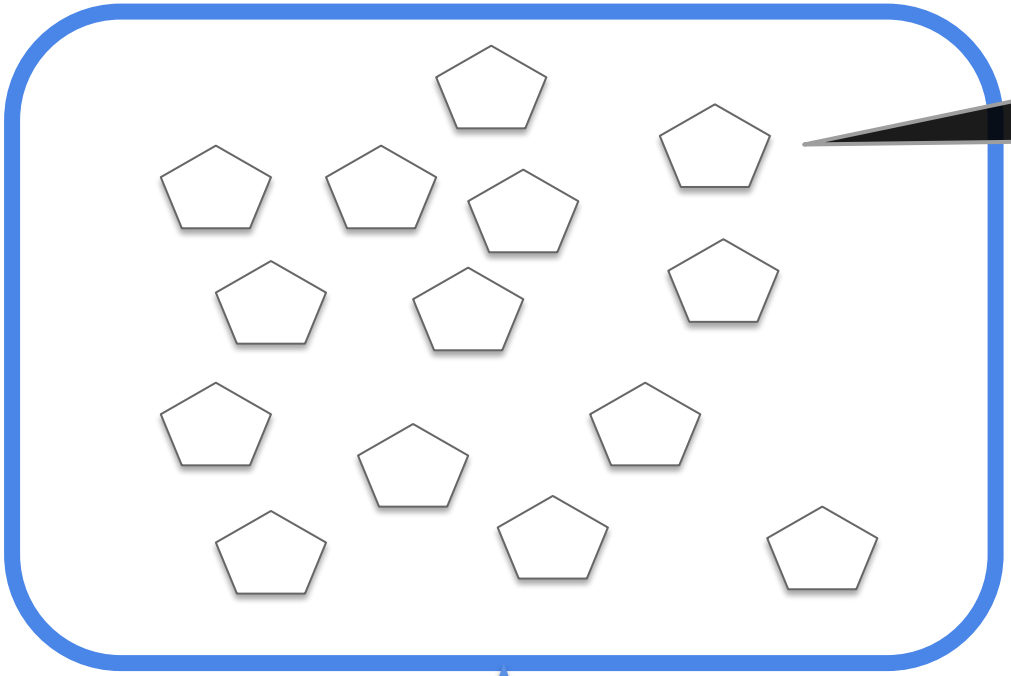
Inference

!

https://forms.gle/9gnrkWD8HkLE5eyc8

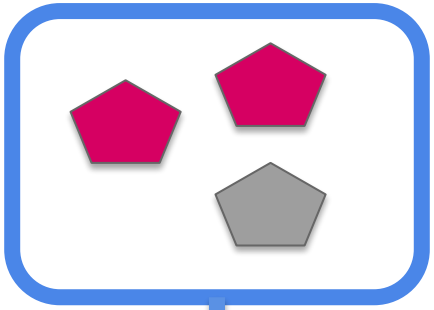
What would you need to consider when sampling air pollution in the US?





Best guess

So we measure pollution levels in a representative sample of US counties

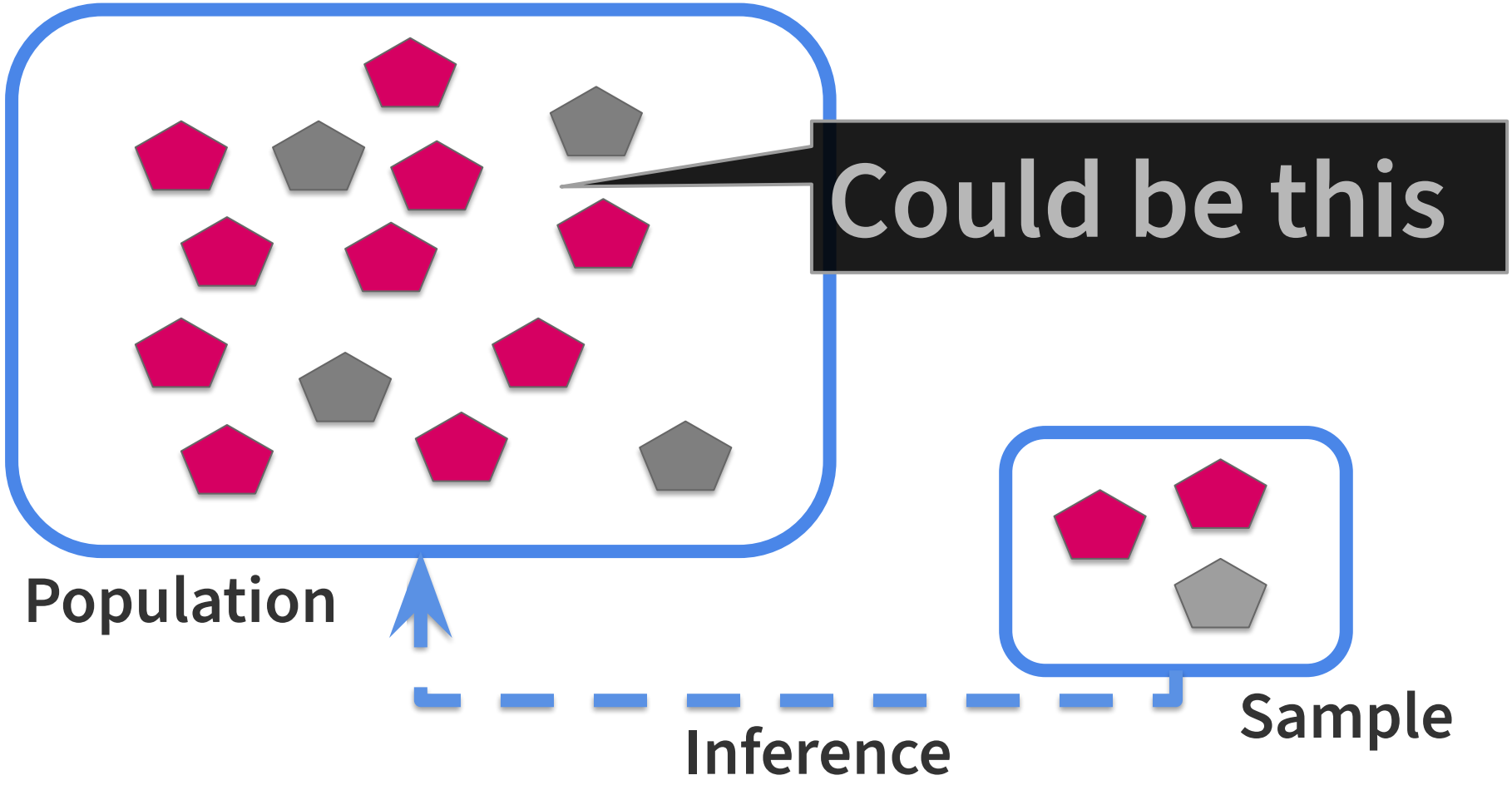


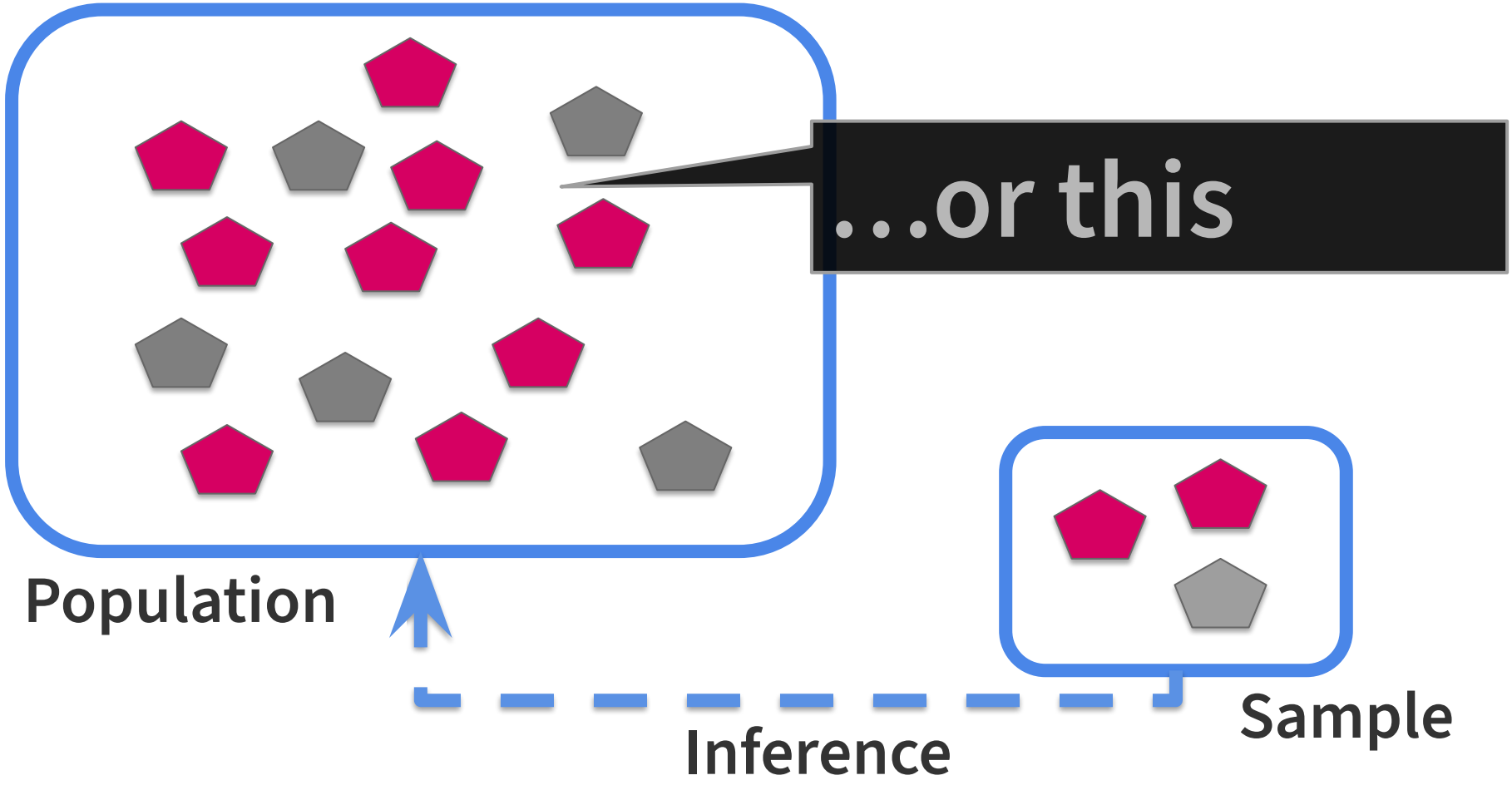
Population

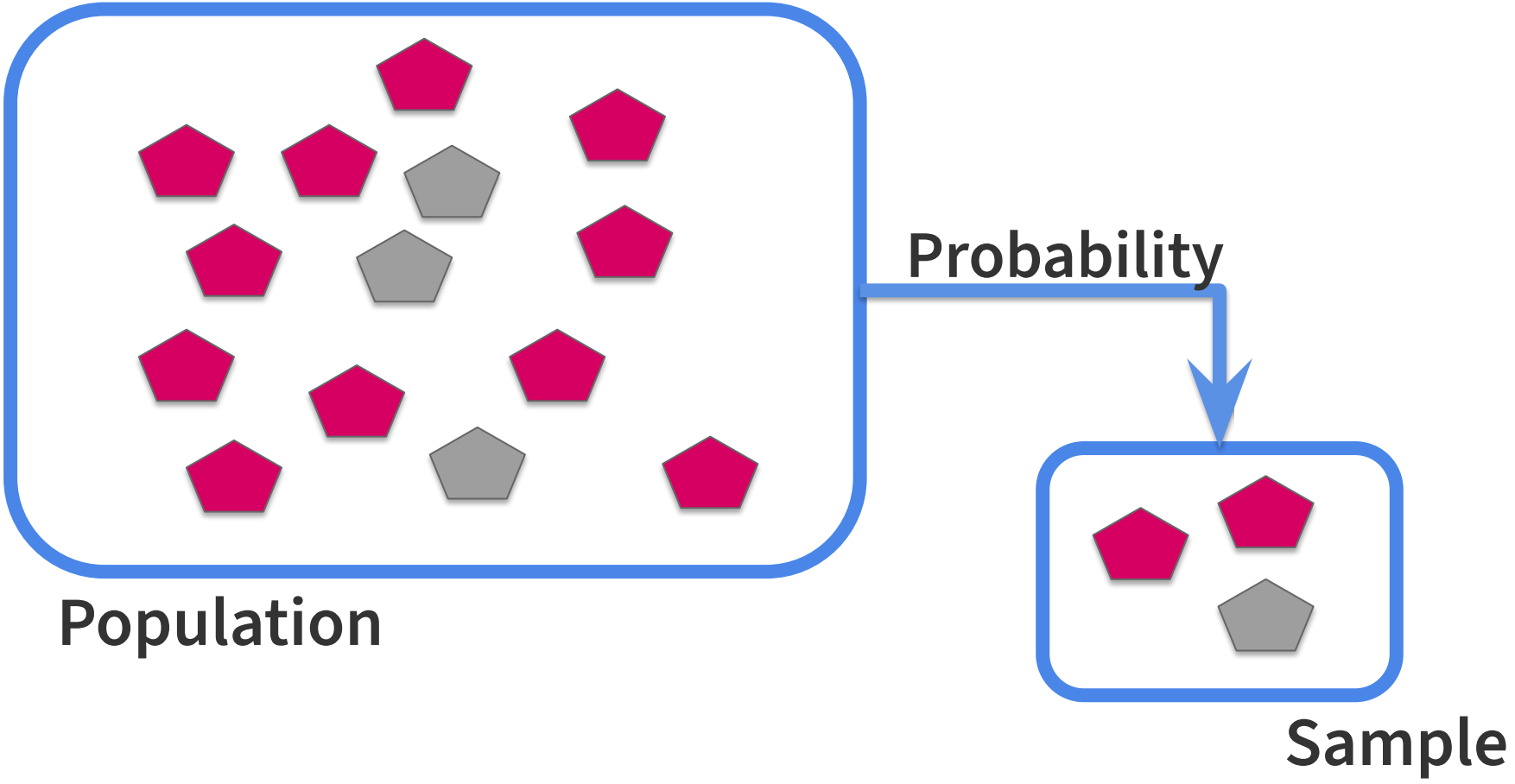
Sample

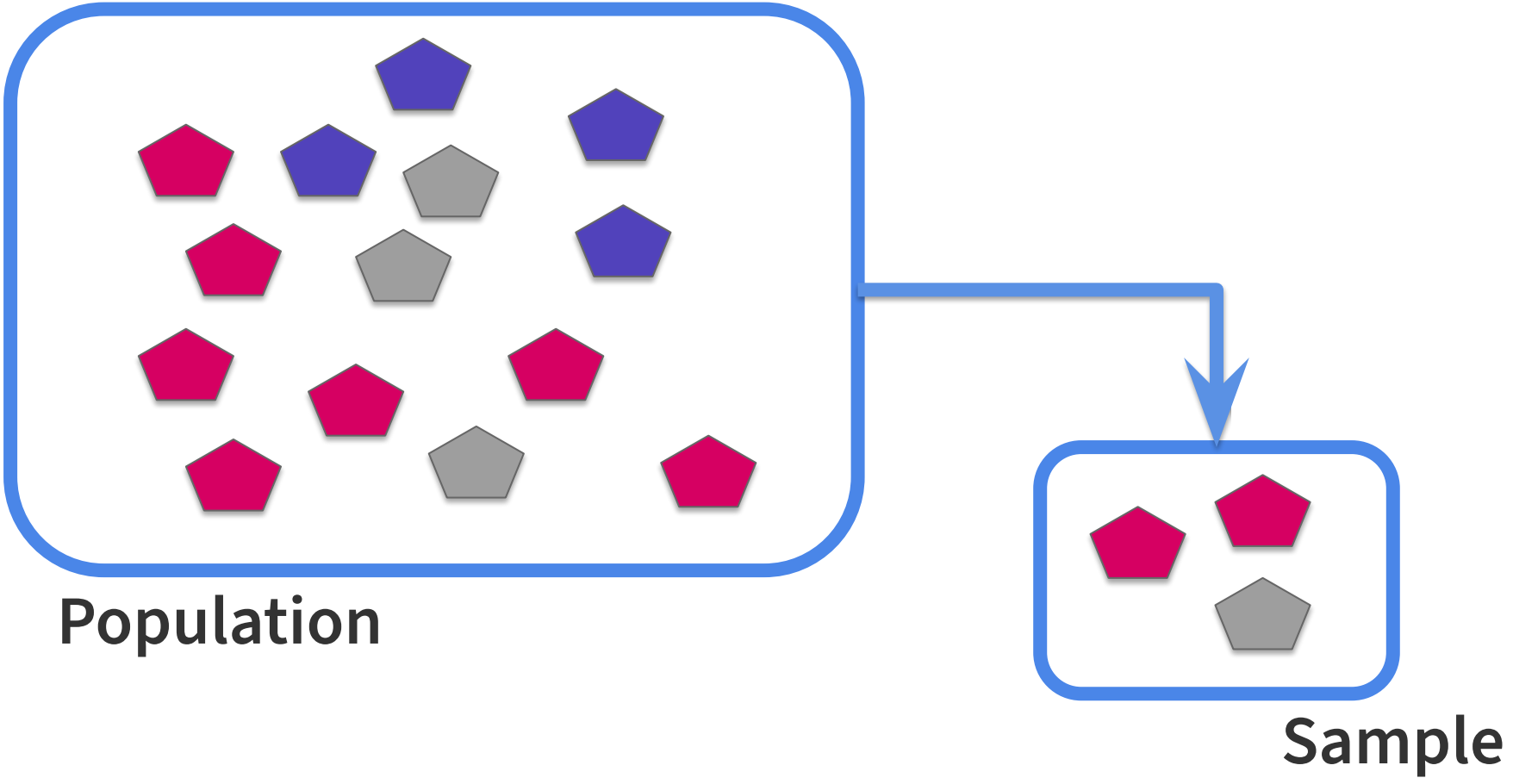
Inference

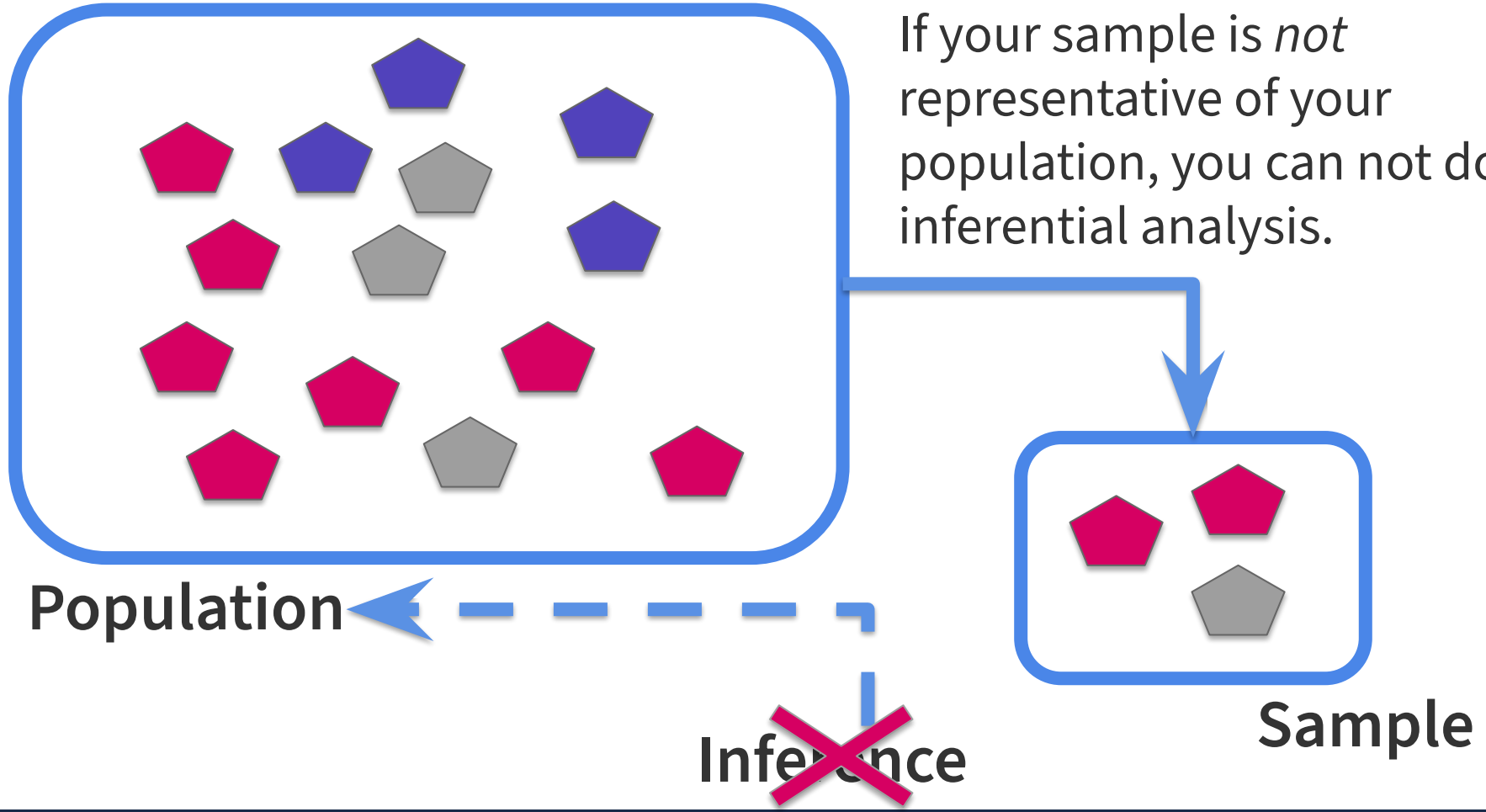
!

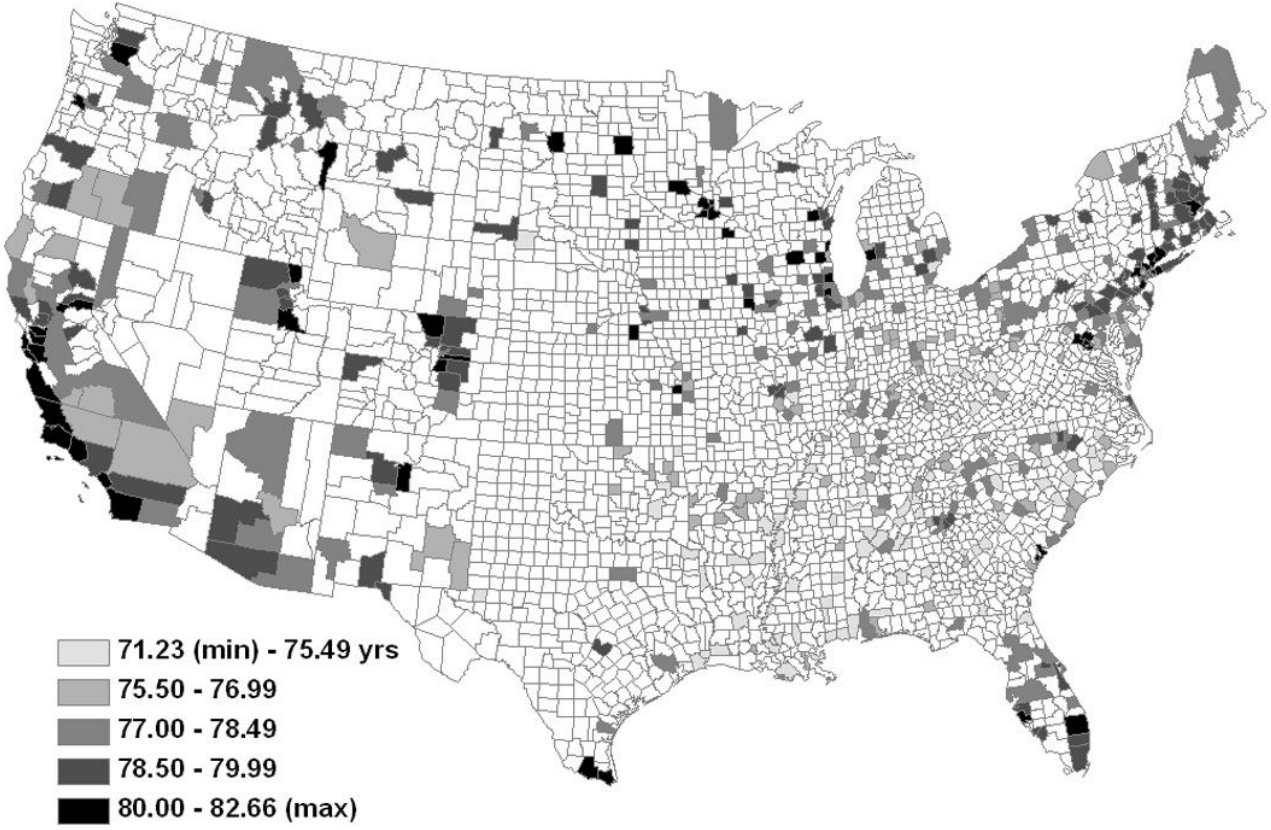

Could be this
Population
Sample
Inference


...or this
Population
Sample
Inference

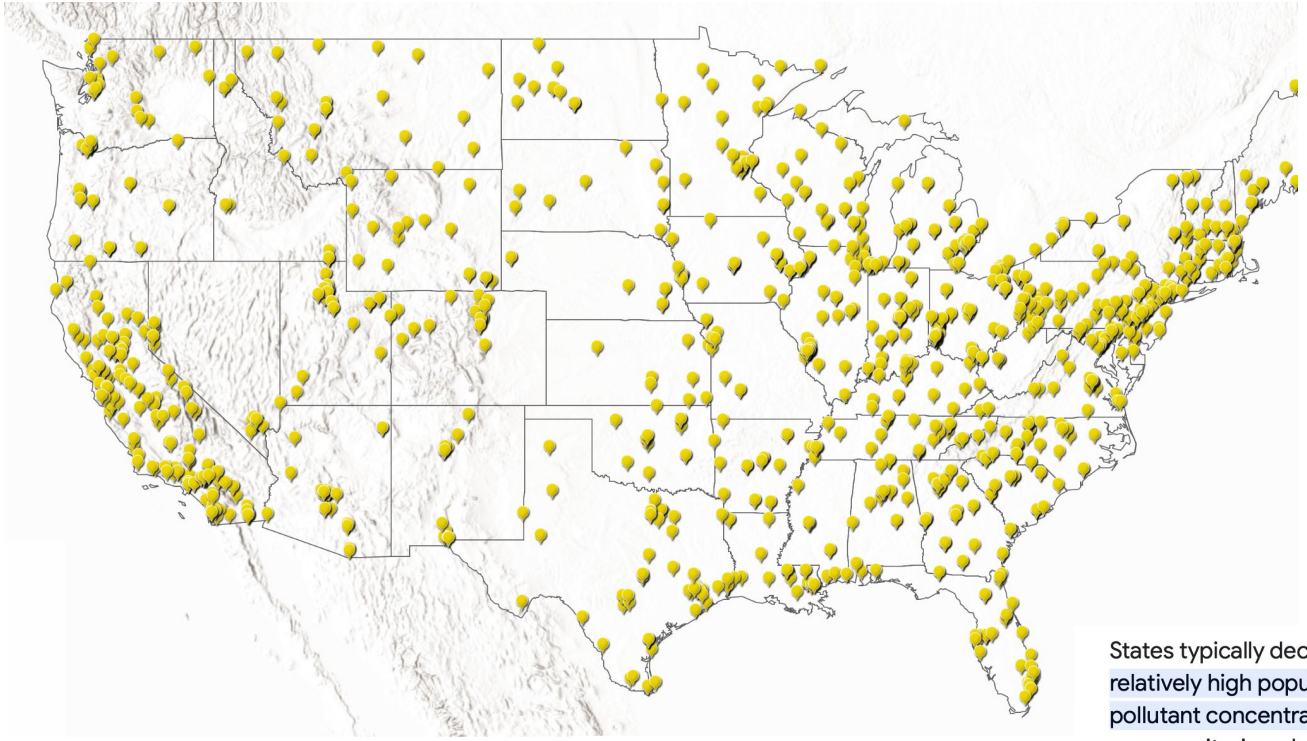

Probability
Population
Sample


Population
Sample


If your sample is *not* representative of your population, you can not do inferential analysis.
Population
Inference
Sample



All counties with with available matching PM2.5 data for 2000 and 2007 from the EPA's Air Quality System. Includes both metropolitan and non-metro counties



States typically decide where monitors are placed based on areas of relatively high population and/or areas believed to have relatively higher pollutant concentrations. Each state is responsible for developing its own monitoring plan, which is then reviewed and revised every five years. Aug 28, 2023

United States Environmental Protection Agency (.gov)
https://www.epa.gov › outdoor-air-quality-data › who-de...

Who decides where monitors get placed? | US EPA

Approaches to Inference

CORRELATION

ASSOCIATION BETWEEN VARIABLES

i.e. Pearson Correlation, Spearman Correlation, chi-square test

COMPARISON OF MEANS

DIFFERENCE IN MEANS BETWEEN VARIABLES

i.e. t-test, ANOVA

REGRESSION

DOES CHANGE IN ONE VARIABLE MEAN CHANGE IN ANOTHER?

I.e. simple regression, multiple regression

NON-PARAMETRIC TESTS

FOR WHEN ASSUMPTIONS IN THESE OTHER 3 CATEGORIES ARE NOT MET

i.e. Wilcoxon rank-sum test, Wilcoxon sign-rank test, sign test

This week: two kinds of questions

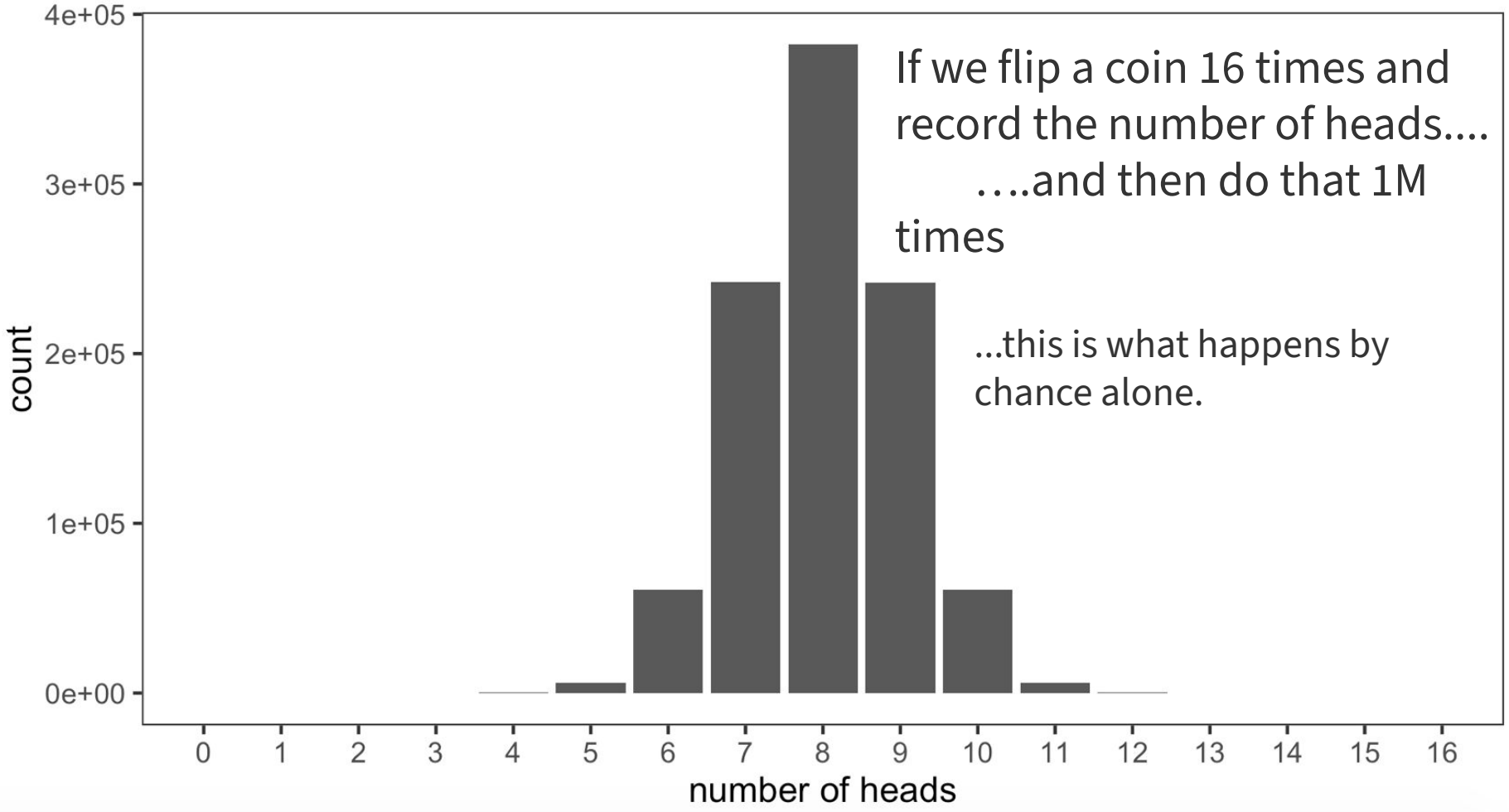
- Fri: Is there a difference between groups?
  - p-values, t-tests, ANOVA
  - Practice in A4 (due Sun, Nov 1)
- Mon: How does one variable change with another?
  - correlation, regression, confounding
  - Practice in A5 (due Sun, Nov 8)

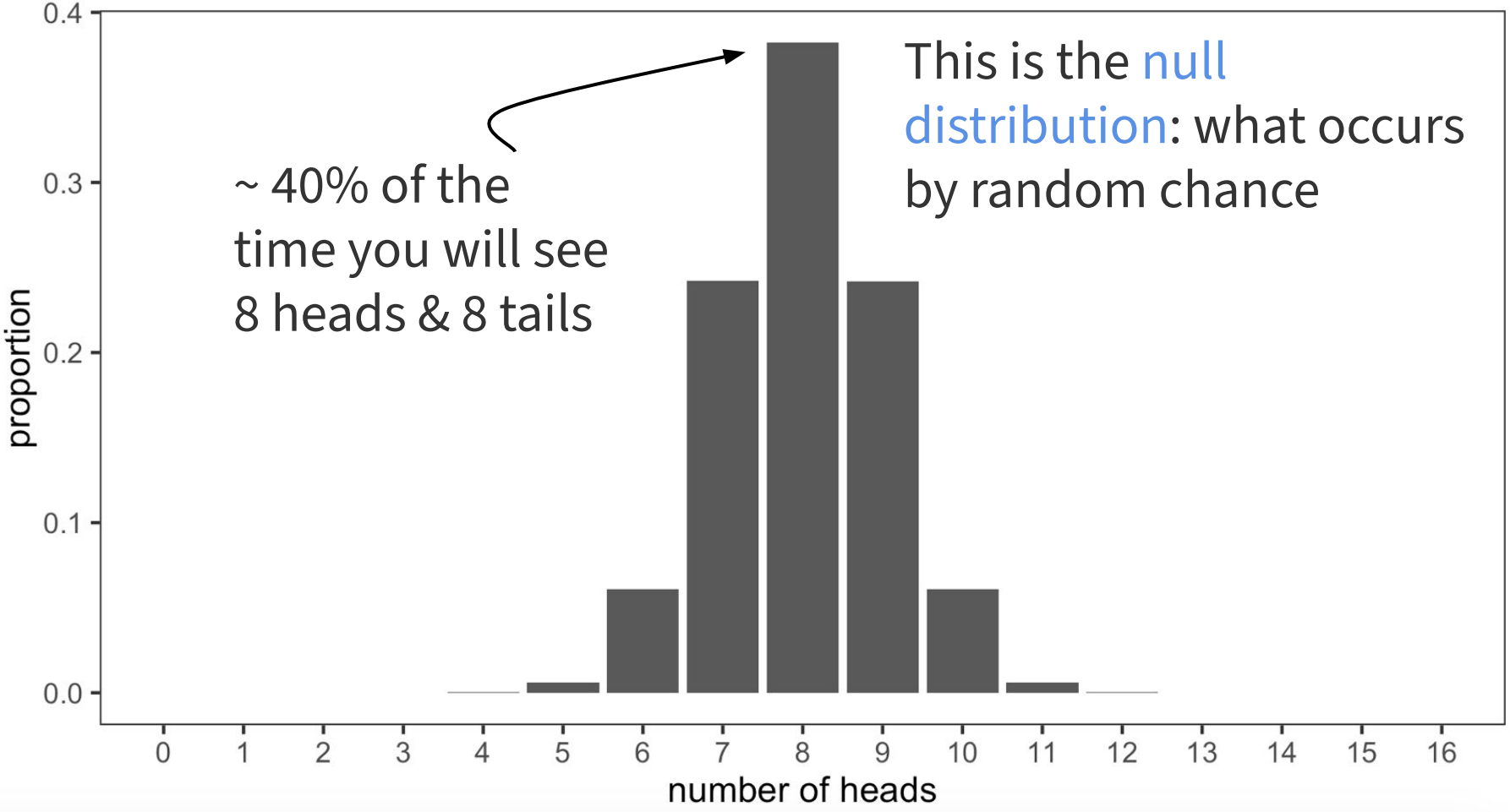
p-value : the probability of getting the observed results (or results more extreme) by chance alone

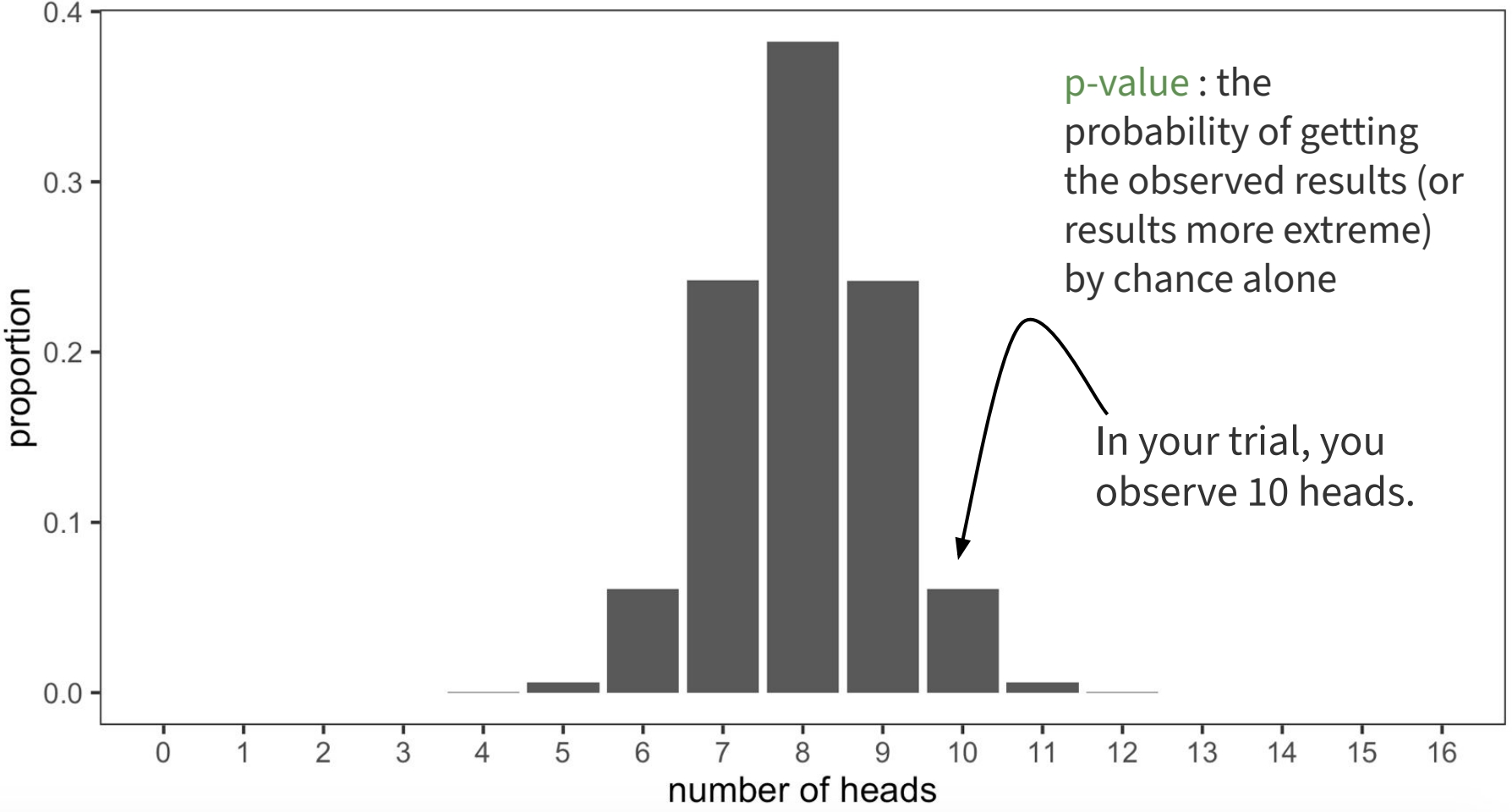
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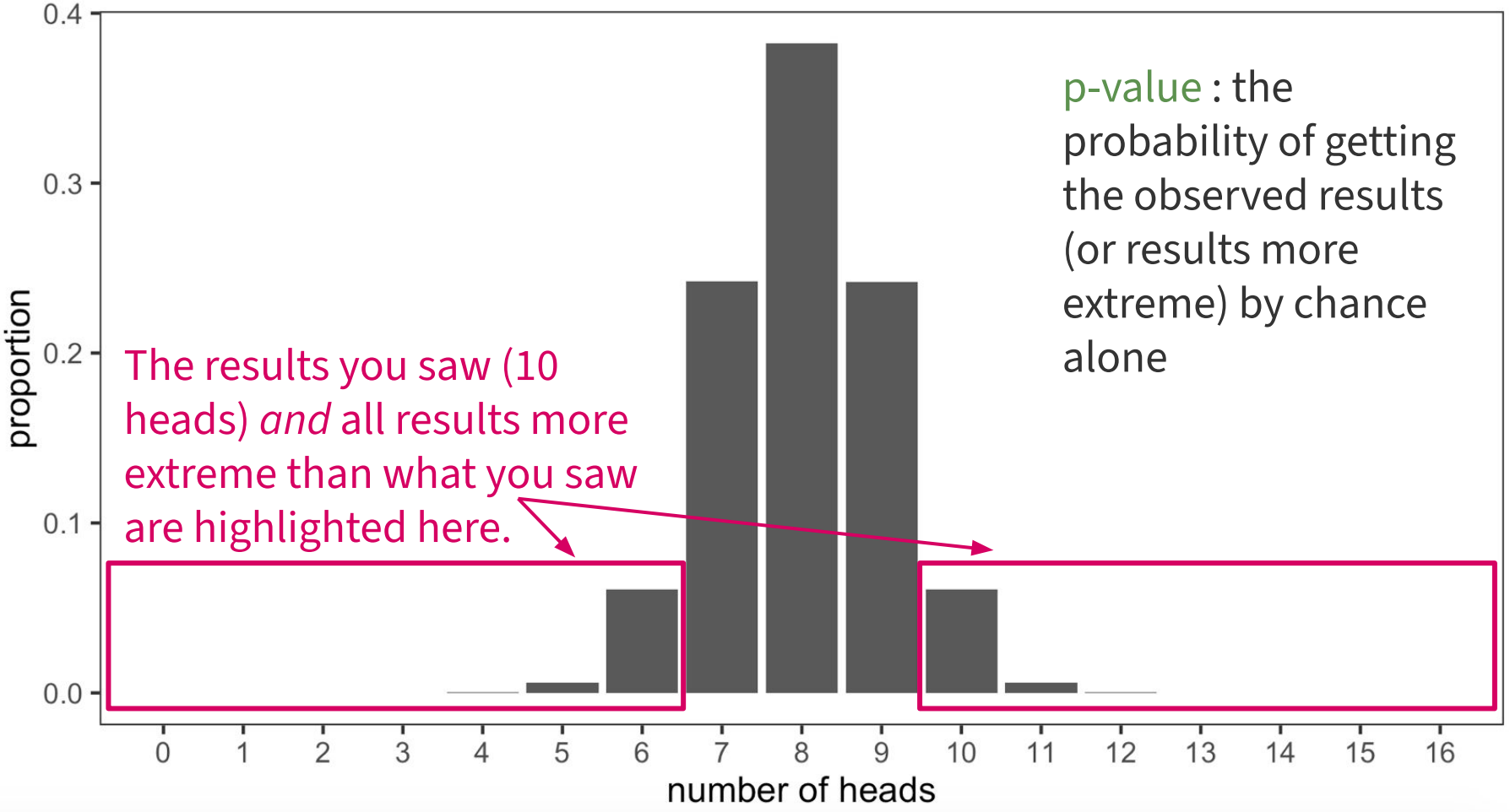


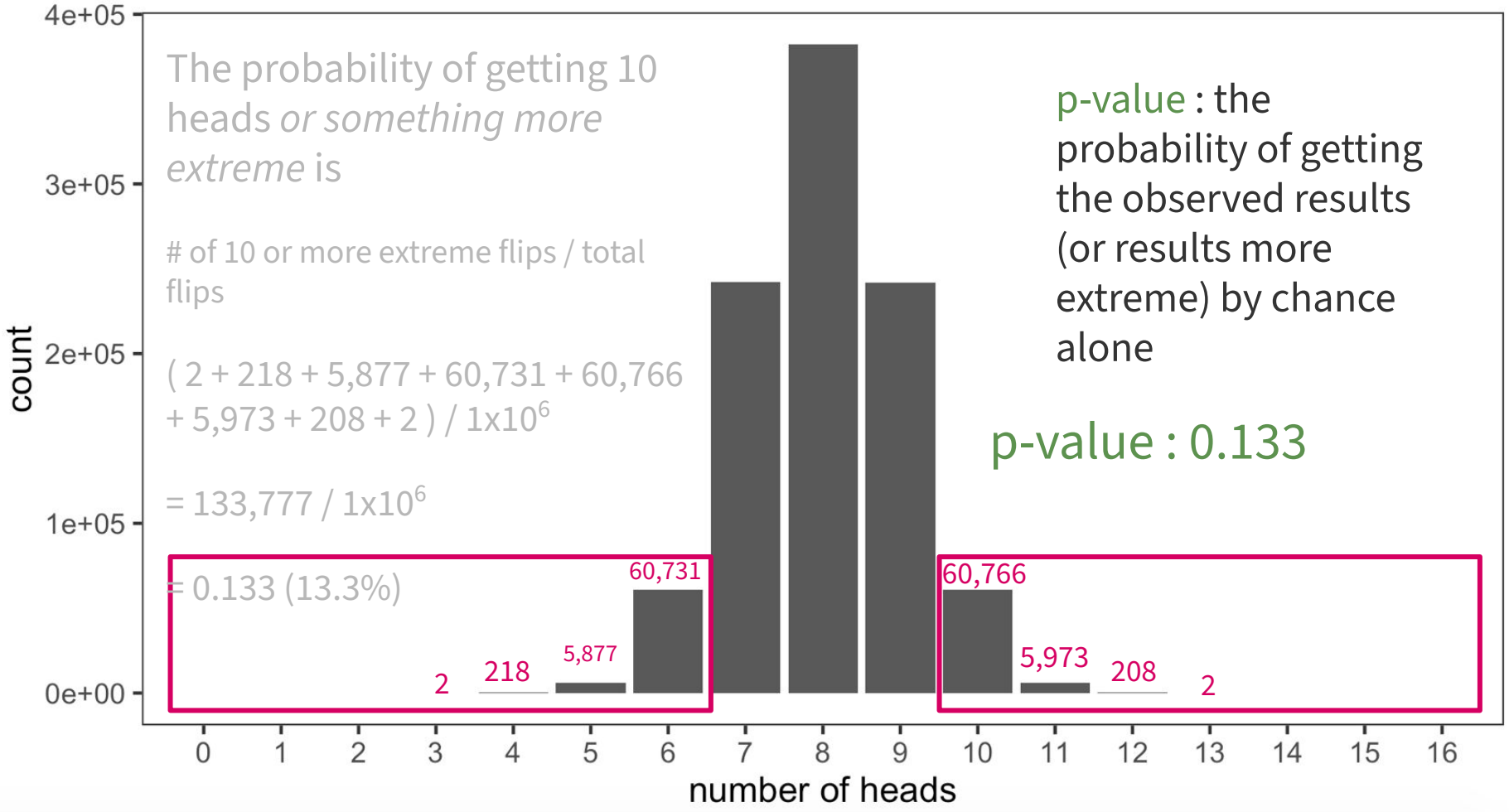
Probability Demo


4e+05
3e+05
2e+05
1e+05
0e+00
count
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
number of heads
If we flip a coin 16 times and record the number of heads....
….and then do that 1M times
...this is what happens by chance alone.

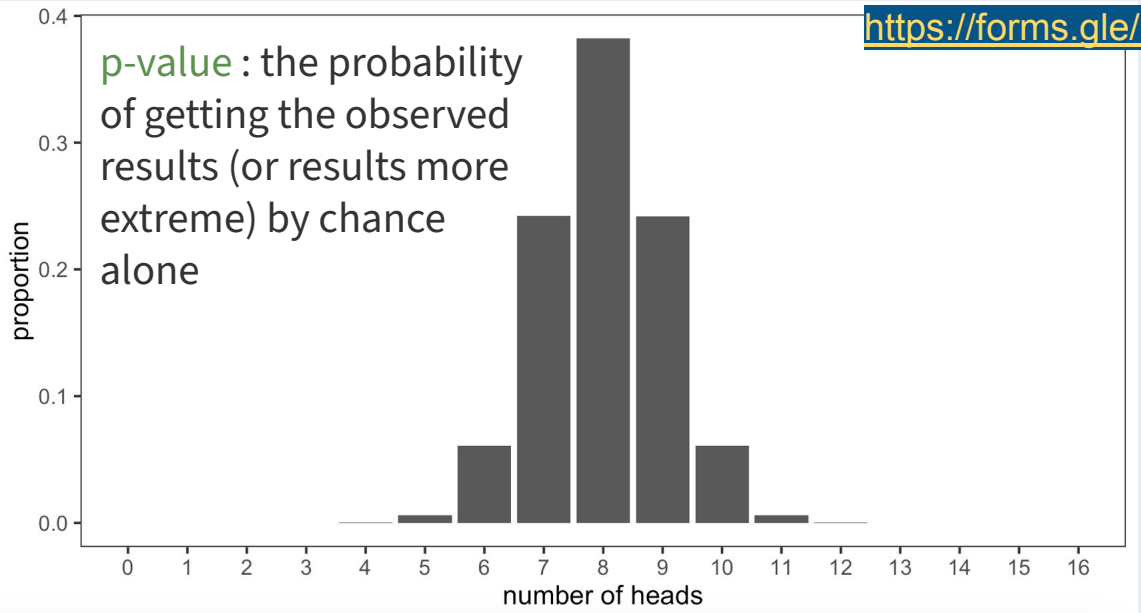

0.4
0.3
0.2
0.1
0.0
proportion
~ 40% of the time you will see 8 heads & 8 tails
This is the null distribution: what occurs by random chance
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
number of heads


0.4
0.3
0.2
0.1
0.0
proportion
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
number of heads
p-value : the probability of getting the observed results (or results more extreme) by chance alone
In your trial, you observe 10 heads.


0.4
0.3
0.2
0.1
0.0
proportion
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
number of heads
p-value : the probability of getting the observed results (or results more extreme) by chance alone
The results you saw (10 heads) *and* all results more extreme than what you saw are highlighted here.


The probability of getting 10 heads *or something more extreme* is
of 10 or more extreme flips / total flips
(2 + 218 + 5,877 + 60,731 + 60,766 + 5,973 + 208 + 2) / 1x10^6
= 133,777 / 1x10^6
= 0.133 (13.3%)
p-value : the probability of getting the observed results (or results more extreme) by chance alone
p-value : 0.133
count
4e+05
3e+05
2e+05
1e+05
0e+00
2
218
5,877
60,731
60,766
5,973
208
2
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
number of heads

https://forms.gle/hVaKuEuUyVEnz4Mv6



What would be the p-value of you flipping 16 heads?

A
< 0.13

B
> 0.13

Simulating the null distribution

- State the null hypothesis (e.g., the coin is fair)
- Simulate the null process many times, computing your statistic each time
- p-value = the fraction of simulations at least as extreme as what you observed
- No formula needed, so it works for almost any statistic
    - You'll see this in today's notebook as a permutation test

p = 0.03. What does that mean?

- A) There's a 3% chance the null hypothesis is true
- B) There's a 97% chance the alternative hypothesis is true
- C) If the null hypothesis were true, results at least this extreme would happen about 3% of the time
- D) The effect is large

Poll question: TODO add to this lecture's form

p < 0.05 is a convention, not a law

- 0.05 = results this extreme happen 1 in 20 times when the null is true
- Run 20 tests on pure noise and you expect about one "significant" result
  - More tests → more false positives
- A p-value is **not** the probability that the null is true
- Report the p-value itself and ***explain what it means for the question you're answering*** (that's more important just "significant" or "not significant")

Significant ≠ big enough to matter

- With enough data, even tiny differences become significant
- So report the effect size with the p-value:
  - Difference in means, with a 95% confidence interval
  - Cohen's d = $(\text{mean}_1 - \text{mean}_2)$ / pooled SD (≈ 0.2 small, 0.5 medium, 0.8 large)
- Then ask: is a difference this big important in the real world?



We're going to the 09-InferenceI Jupyter notebook now

Comparing means

t-test: comparing two means

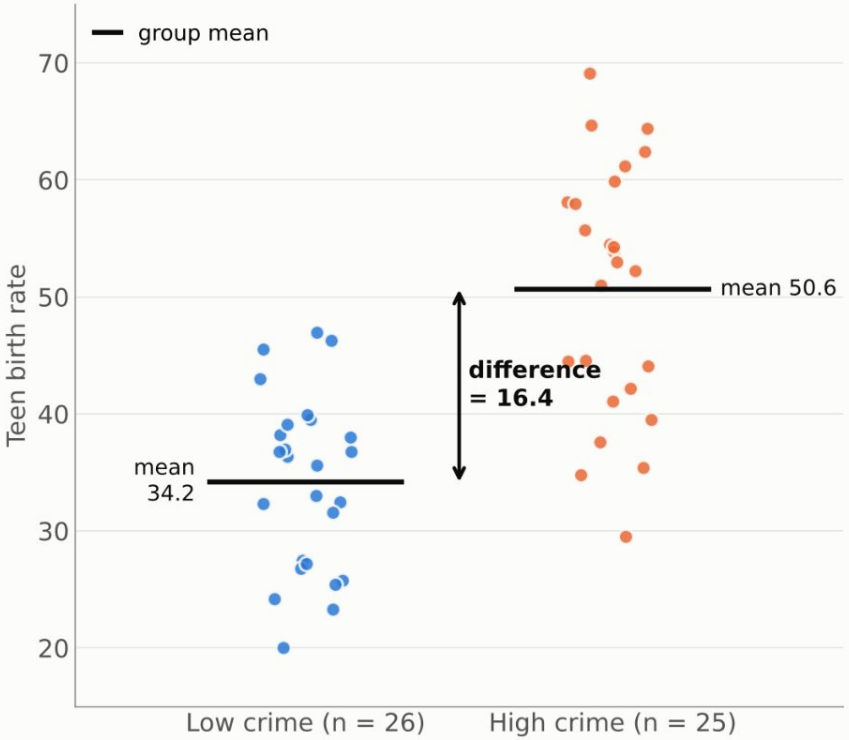
- $H_0: \mu_1 = \mu_2$ $H_a: \mu_1 \neq \mu_2$
- t = difference in means ÷ its standard error
  - Bigger difference → larger t
  - More spread or smaller samples → smaller t
- Today's notebook: states with above-median violent crime have teen birth rates 16 per 1,000 higher (95% CI: 11 to 22; $p < 0.001$)

Check the assumptions before you run a t-test

- Continuous outcome
- Approximately normal in each group
  - Check with a QQ plot or stats.normaltest()
- Equal variances in each group
  - If not: Welch's t-test, ttest_ind(..., equal_var=False)
- Independent observations
  - Same subjects measured twice? Paired t-test, ttest_rel()

t-test: is the difference between two means bigger than chance?

Teen birth rate (births per 1,000) in states with below- vs. above-median violent crime



1. Check the assumptions

✓ Continuous outcome, independent states (not paired)

✓ Roughly normal in each group (normaltest p = 0.54, 0.40)

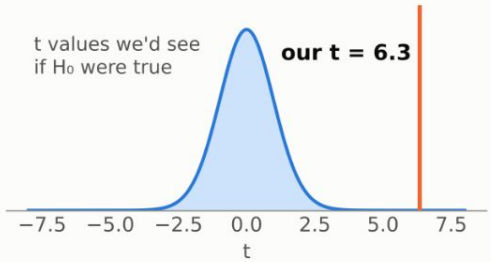
! Variances differ about 2x (56 vs. 115), so use Welch's t-test

2. The test statistic

$$t = \frac{\text{difference in means}}{\text{standard error of the difference}}$$

Bigger difference → larger t. More spread or smaller samples → smaller t.

3. Compare t to what chance alone produces



Result

p < 0.001

Difference: 16.4

95% CI: 11.2 to 21.7

Cohen's d = 1.8 (large)

H_0: the two population means are equal. H_a: they differ. Report the difference and its CI, not just the p-value.

Same subjects measured twice? Paired t-test (stats.ttest_rel). Not normal or ordinal? Mann-Whitney U (stats.mannwhitneyu).

Data: Mind On Statistics (Utts & Heckard). Groups split at the median violent crime rate.

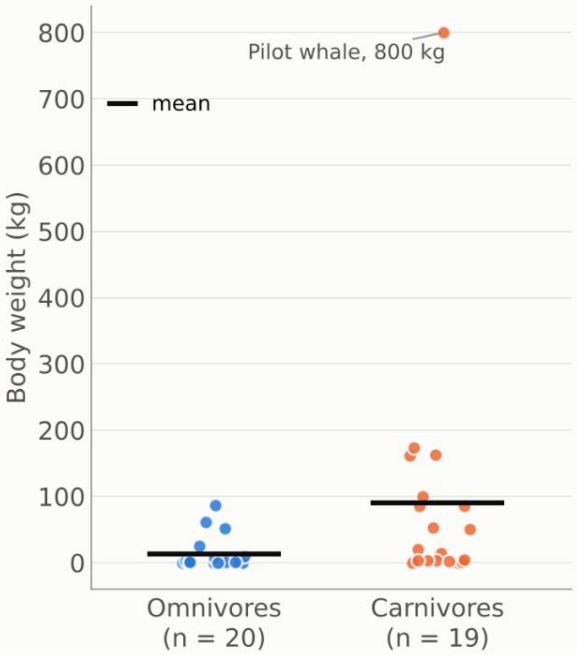
When the data aren't normal: rank-based tests

- Replace each value with its rank, then compare ranks across groups
  - Two groups: **Mann-Whitney U** (aka Wilcoxon rank-sum)
  - Three or more groups: **Kruskal-Wallis**
- Useful for skewed or ordinal data (e.g., ratings from 1 to 5)
- Cost: less power than a t-test when the t-test's assumptions do hold
- Another option: transform the data (e.g., log) so they're closer to normal

Mann-Whitney U: compare two groups by rank, not by value

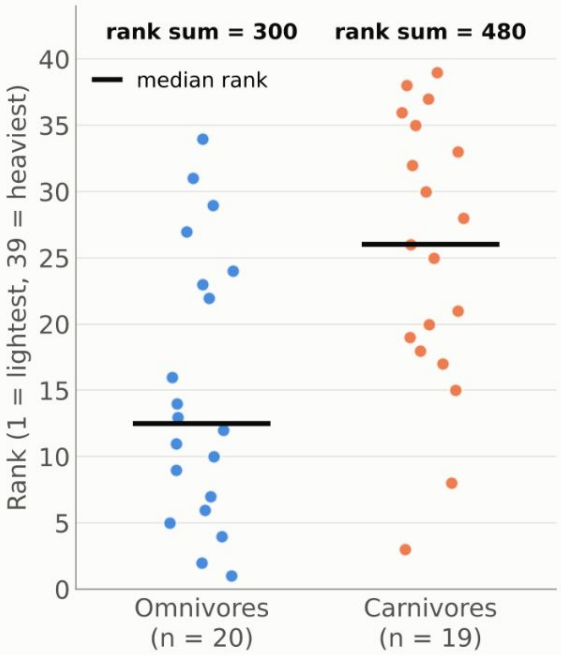
Body weight of carnivores vs. omnivores (mammal sleep data)

A. The raw values



Very skewed: a few huge animals pull the carnivore mean up and swamp everything else.

B. The same animals, ranked



The pilot whale is now just rank 39. No single value can dominate.

How the test works

1. Pool both groups and rank every value
2. Add up the ranks in each group
3. U = rank sum − $n(n + 1)/2$

   Carnivores: $480 - 19 \cdot 20/2 = 290$

U runs from 0 to $n_1 \cdot n_2 = 380$. If neither group tends to be larger, $U \approx 190$.

Result

Mann-Whitney U = 290, p = 0.005

Carnivores tend to be heavier.

Welch's t-test on the same data: p = 0.08

The outliers inflate the spread, so the t-test misses a difference the ranks show.

H_0: values in one group are no more likely to be larger than values in the other. Use for skewed data, outliers, or ordinal data (e.g., ratings).

Cost: less power than a t-test when the t-test's assumptions hold. In Python: stats.mannwhitneyu(group1, group2). 3+ groups: Kruskal-Wallis.

Data: msleep (ggplot2), from Savage & West (2007). Animals with a known diet only.

ANOVA: comparing more than two means

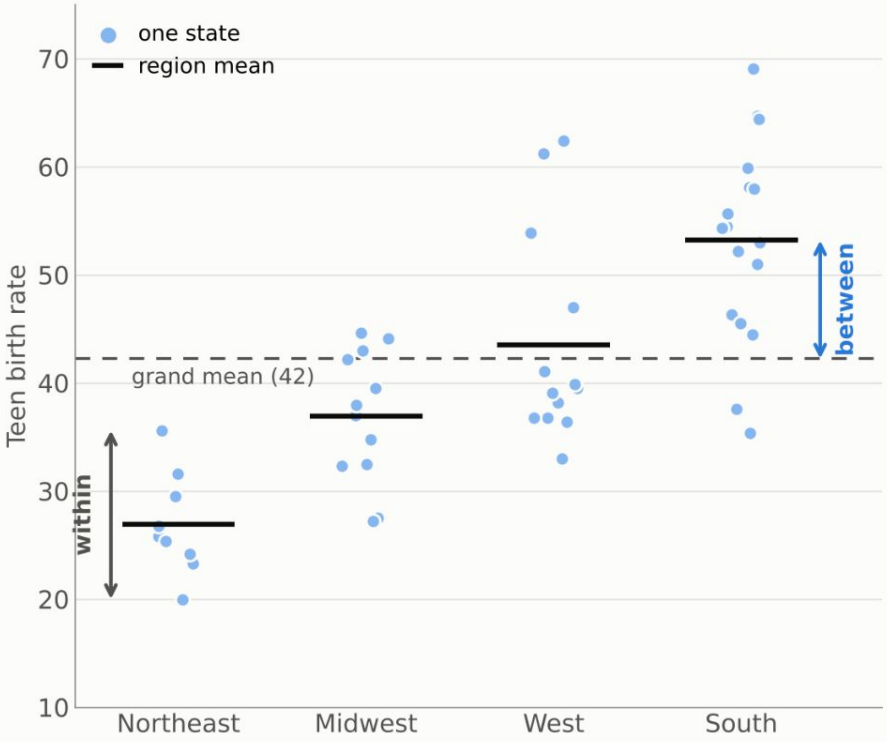
- 4 regions → 6 pairwise t-tests, each another chance at a false positive
- One-way ANOVA asks a single question:
  - H_0: all group means are equal H_a: at least one mean differs
- F = variation between group means ÷ variation within groups
- Same assumptions as the t-test (Welch's ANOVA if variances differ)

Tukey's HSD: which groups differ?

- A *post-hoc* test: compares every pair of groups
- Adjusts p-values for the number of comparisons, so the overall false-positive rate stays at 5%
- Each row of the output: difference in means, adjusted p-value, 95% CI, reject (True/False)
- In today's notebook: teen birth rate by US Census region

One-way ANOVA: is at least one group mean different?

Teen birth rate (births per 1,000) for 50 US states + DC, by Census region



1. Is any mean different? The F-test

$$F = \frac{\text{variation between group means}}{\text{variation within groups}}$$

Big F means far apart compared with the noise → small p

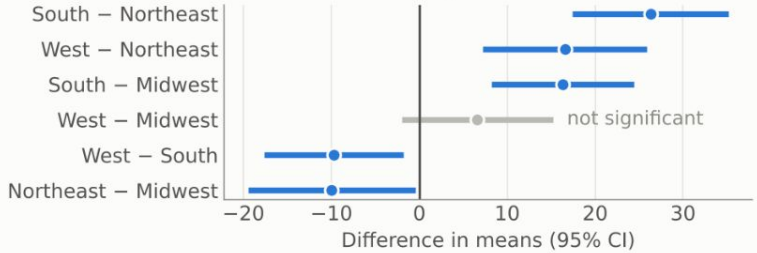
F near 1 differences in means look like noise → large p

H_0: all group means are equal H_a: at least one is different

Here: F = 23.0, p < 0.001, so at least one region differs.

2. Which pairs differ? Tukey's HSD

Compares every pair, adjusting p-values for the number of comparisons.
A confidence interval that crosses 0 means no significant difference.



Assumptions (same as the t-test): approximately normal within each group, equal variances, independent observations.

Unequal variances? Welch's ANOVA, the default in sm.stats.anova_oneway (here F = 31.6, p < 0.001). Not normal? Kruskal-Wallis (stats.kruskal).

Data: Mind On Statistics (Utts & Heckard); US Census Bureau regions.

Which test should you use?

Comparing…	Assumptions met	Assumptions not met
2 independent groups	t-test (Welch's if variances differ)	Mann-Whitney U
2 paired measurements	Paired t-test	Wilcoxon signed-rank
3+ groups	ANOVA, then Tukey's HSD	Kruskal-Wallis